JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY **9648/01**

Paper 1 Multiple Choice **20 September 2013**

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and class on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

This document consists of **19** printed pages.

[Turn over

1 Which row correctly identifies all the locations of ribosomes in a eukaryotic cell?

	free in cytoplasm	in mitochondria	attached to endoplasmic reticulum	attached to nuclear envelope	in nucleus
A	✓	✗	✓	✗	✗
B	✓	✓	✓	✗	✗
C	✗	✓	✓	✗	✗
D	✗	✗	✓	✓	✓

2 What are the features of triglycerides?

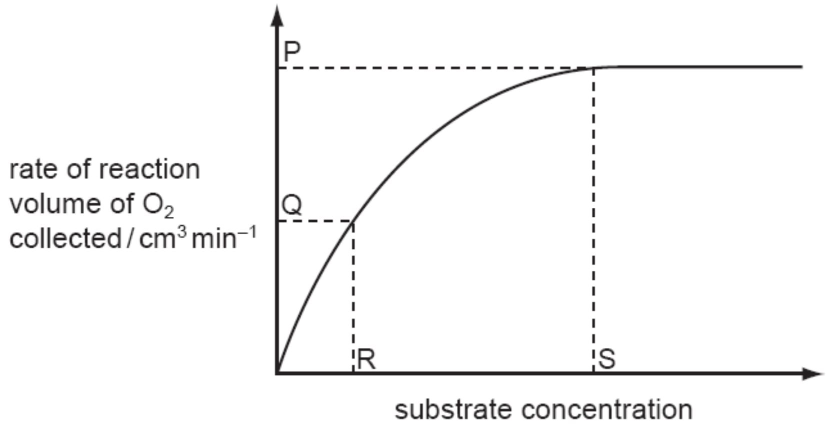
	polar	less dense than water	higher energy value than carbohydrates	lower proportion of hydrogen than in carbohydrates
A	✓	✓	✗	✗
B	✓	✗	✓	✓
C	✗	✓	✓	✗
D	✗	✗	✗	✓

3 Liver tissue produces an enzyme called catalase which breaks down hydrogen peroxide into water and oxygen.

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time.

Students added small cubes of fresh liver tissue to a range of hydrogen peroxide solutions and measured the volumes of oxygen produced. Their data were used to produce the graph showing how changing the concentration of hydrogen peroxide affected the rate of oxygen production.



Which statements are correct?

1 At P, the rate of reaction is limited by the concentration of enzyme.
2 At Q, all of the enzyme active sites are occupied by substrate molecules.
3 At Q, the rate of reaction is limited by the concentration of the substrate.
4 R represents K_m where the reaction rate = V_{max} / 2.
5 At S, all of the enzyme active sites are occupied by substrate molecules.

A 2 and 3 only **B** 1, 2, 3 and 4 only **C** 1, 2, 4 and 5 only **D** 1, 3, 4 and 5 only

4 Which statement is incorrect for mitotic cell division?

A DNA is replicated semi-conservatively during mitosis.
B DNA is normally unchanged from one generation of cells to the next.
C The daughter cells have the potential to produce the same enzymes as the parent cell.
D The same quantity of DNA is distributed to the nuclei of two new cells.

5 For organisms undergoing sexual reproduction, a reduction division occurs before fertilisation.

Which reason(s) explain why this is necessary?

1 increase genetic variation
2 prevent doubling of the chromosome number
3 reduce the chances of mutation

A 1 only **B** 2 only **C** 1 and 2 only **D** 1, 2 and 3

6 Which features of DNA enable it to meet these requirements as a molecule of inheritance?

	requirement of DNA molecule			
	ability to remain stable	ability to contain information	ability to transfer information	ability to replicate
A	complementary base pairing	formation of mRNA for translation	sequences of nucleotides	sugar-phosphate backbone
B	formation of mRNA for translation	complementary base pairing	sugar-phosphate backbone	sequences of nucleotides
C	sequences of nucleotides	sugar-phosphate backbone	complementary base pairing	formation of mRNA for translation
D	sugar-phosphate backbone	sequences of nucleotides	formation of mRNA for translation	complementary base pairing

7 Which statements about the genetic code are correct?

1 The genetic code has redundancy and is degenerate.
2 There is only one codon for the amino acid methionine.
3 Codons act as 'stop' and 'start' signals during transcription and translation.
4 Prokaryotes generally use the same genetic code as eukaryotes.

A 1 and 2 only **B** 1, 2 and 3 only **C** 1, 2 and 4 only **D** 1, 2, 3 and 4

8 Ribonuclease is an enzyme that digests RNA. The first five amino acids of the functional molecule of ribonuclease are:

lys – glu – thr – ala – ala

The mRNA of the gene coding for ribonuclease, for the first 15 nucleotides, has the following sequence.

AUGAAGGAAACUGCU

The genetic code table is shown below.

first position	second position U	C	A	G	third position
U	phe phe leu leu	ser ser ser ser	tyr tyr STOP STOP	cys cys STOP trp	U C A G
C	leu leu leu leu	pro pro pro pro	his his gln gln	arg arg arg arg	U C A G
A	ile ile ile met	thr thr thr thr	asn asn lys lys	ser ser arg arg	U C A G
G	val val val val	ala ala ala ala	asp asp glu glu	gly gly gly gly	U C A G

Which event occurs to explain the information given above?

A The first amino acid on the polypeptide chain is removed in post-translational modification.
B The first codon is removed from the mRNA transcript in post-transcriptional modification.
C The mRNA binds to the rRNA in the second codon position.
D There is no tRNA with an anticodon complementary to the first codon.

9 Base substitution mutations can have the following molecular consequence except

A changes one codon for an amino acid into another codon for that same amino acid.
B codon for one amino acid is changed into a codon of another amino acid.
C reading frame changes downstream of the mutant site.
D codon for one amino acid is changed into a translation termination codon.

10 What happens when a bacterial cell that is infected by lambda phage divides?

A Viral RNA is replicated.
B Viral RNA is transcribed from viral DNA.
C Viral DNA is replicated.
D Viral DNA is made from viral RNA.

11 The following describes events in the life cycle of an influenza virus. Which statement is correct?

A Acidification results in the viral envelope fusing with the exocytic vesicle membrane, releasing the capsid in the process.
B Viral genome acts as a template for the making of new copies of viral RNA genome.
C Complementary RNA strand is translated into viral proteins.
D Glycoproteins synthesised by ribosomes are transported to the plasma membrane via endocytic vesicles whereby the glycoproteins are incorporated into the host cell's plasma membrane.

12 A scientist is using an ampicillin-resistant strain of bacteria which has a non-functional *lac*Z gene.

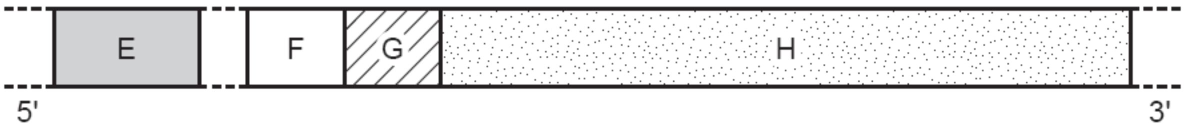
She has a F plasmid containing a functional copy of *lac*Z gene. She then adds a high concentration of the plasmid to a tube of the bacteria in a medium for bacterial growth that contains glucose or lactose as the only energy source. This tube (+) and a control tube (-) with similar bacteria but no plasmid are both incubated under the appropriate conditions for growth and plasmid uptake. The scientist then spreads a sample of each bacterial culture (+ and -) on each of the three types of plates indicated below.

Plate A: Glucose medium
Plate B: Glucose medium with ampicillin
Plate C: Lactose medium with ampicillin

Which row correctly identifies the plates with growth (✓) or no growth (×)?

	Bacterial strain with added plasmid (+)			Bacterial strain with no plasmid (-)		
	Plate A	Plate B	Plate C	Plate A	Plate B	Plate C
A	✓	✓	✓	✓	✓	✓
B	✓	✓	✓	✓	✓	×
C	✓	×	×	×	×	×
D	×	×	✓	×	×	×

13 The diagram represents a length of DNA which forms a structure called an operon. Parts of the operon are labelled E, F, G and H. They have different functions.



What identifies the functions of parts E, F, G and H?

	E	F	G	H
A	operator	structural gene(s)	regulator	promoter
B	promoter	regulator	structural gene(s)	operator
C	regulator	promoter	operator	structural gene(s)
D	structural gene(s)	operator	promoter	regulator

14 Which statement correctly describes the role of histone proteins?

A All eukaryotic genes are transcribed continuously because they are not packaged by histones.
B A DNA must be selectively released from its histone packaging before transcription can occur in bacteria.
C Histones package prokaryote chromatin into the nucleosomes that form the bulk of the chromosome.
D The organisation of DNA by histones in eukaryotes allows some gene control sequences to be thousands of base pairs away from the gene concerned.

15 In transcriptional control in eukaryotic cells

1 a different combination of DNA binding proteins may regulate the activity of a particular gene.
2 enhancers may be involved in the promotion as well as regulation of gene transcription.
3 phosphorylation of transcriptional factors by a kinase may occur.
4 enhancers may be some distance from the promoter sites they control.

A 1, 2 and 3 only **B** 1, 3 and 4 only **C** 2, 3 and 4 only **D** 1, 2, 3 and 4

16 Which of the following is least likely to cause a proto-oncogene to become an oncogene?

A A gene is incorporated into a retroviral genome.
B A gene is moved close to an enhancer, causing excess product to be made.
C A gene is truncated, yielding a protein with modified activity.
D A gene is moved into centromeric heterochromatin, silencing its transcription.

17 A student has 2 cultures of cells. Culture A consists of normal cells, while culture B consists of cells which were isolated from a tumour and divide uncontrollably. He wants to determine if the uncontrolled growth of cells in culture B was the result of activation of an oncogene or the mutation of a tumour suppressor gene. He decides to fuse the cells from the two cultures to form a hybrid line containing DNA from both cells.

The list of possible results and the conclusions drawn from the experiment are summarised by the student in the table below:

	resulting growth of hybrid cells	type of gene mutated in the tumour
1	grows normally	activated oncogene
2	grows uncontrollably	activated oncogene
3	grows normally	mutated tumour suppressor gene
4	grows uncontrollably	mutated tumour suppressor gene

Only two of the conclusions are correctly matched with the possible results. The two correct possibilities are

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

18 A parent organism of unknown genotype is mated in a test cross. Half of the offspring have the same phenotype as the parent.

What can be concluded from this result?

A The parent is heterozygous for the trait.
B The parent is homozygous dominant for the trait.
C The parent is homozygous recessive for the trait.
D The trait being inherited is polygenic.

19 The table shows the results of a series of crosses in a species of small mammal.

coat colour phenotype		
male parent	female parent	offspring
dark grey	light grey	dark grey, light grey, albino
light grey	albino	light grey, white with black patches
dark grey	white with black patches	dark grey, light grey
light grey	dark grey	dark grey, light grey, white with black patches

What explains the inheritance of the range of phenotypes shown by these crosses?

A one gene with a pair of co-dominant alleles
B one gene with multiple alleles
C sex linkage of the allele for grey coat colour
D two genes, each with a dominant and recessive allele

20 Two genes A and B are linked together as shown below.

$$\frac{\underline{A \quad b}}{a \quad B}$$

If the genes are far enough apart such that crossing over between the alleles occurs occasionally, which statement is true of the gametes?

A All of the gametes will be Ab and aB.
B There will be 25% Ab, 25% aB, 25% ab and 25% AB.
C There will be approximately equal numbers of Ab and ab gametes.
D The number of Ab gametes will be greater than the number of ab gametes.

21 In the fruit fly, *Drosophila melanogaster*, four genes whose recessive alleles code for black body (B / b), curved wings (C / c), purple eyes (P / p) and vestigial wings (V / v) are linked on chromosome 2.

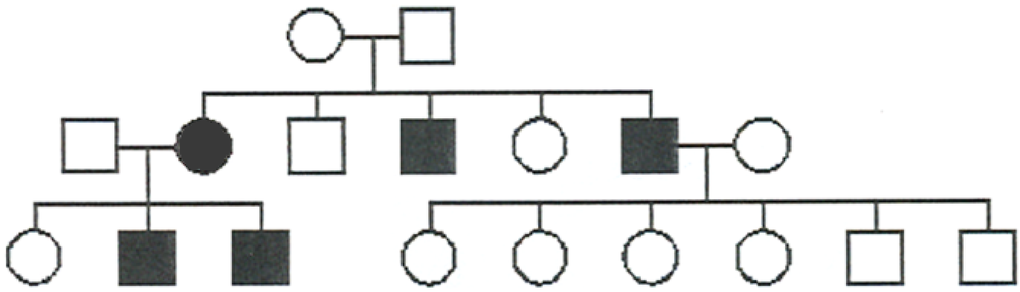
The table shows some distances between the gene loci, as determined by breeding experiments.

gene loci	distance between loci / map units
B / b and P / p	6.0
B / b and V / v	18.5
P / p and V / v	12.5
P / p and C / c	21.0
V / v and C / c	8.5

What is the correct sequence of the loci on chromosome 2?

A	B / b	P / p	V / v	C / c
B	C / c	V / v	B / b	P / p
C	P / p	V / v	C / c	B / b
D	V / v	B / b	P / p	C / c

22 The following pedigree depicts the inheritance of a rare hereditary disease affecting muscles.



What is the mode of inheritance of this disease?

A autosomal dominant
B autosomal recessive
C X-linked dominant
D X-linked recessive

23 The phenotypes of 200 offspring of a dihybrid test cross were recorded. The cross involved petal colour and fertility of the anthers of sweet pea flowers. The table shows the observed and expected numbers of each phenotype.

phenotype	purple petals fertile anthers	purple petals sterile anthers	maroon petals fertile anthers	maroon petals sterile anthers
observed numbers	87	14	16	83
expected numbers	50	50	50	50

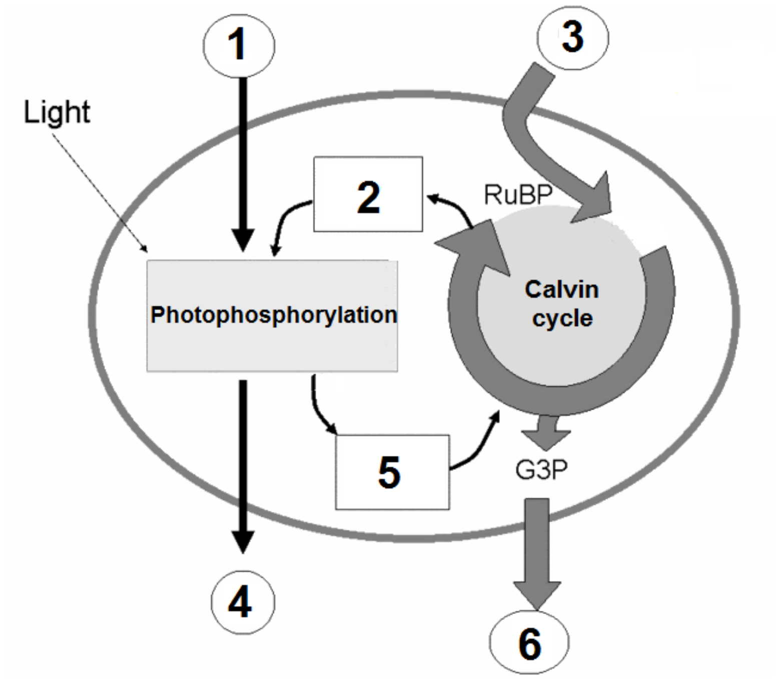
A chi-squared (χ^2) test was performed and the probability of the difference between the observed and expected results being due to chance was found to be <0.001.

Which conclusions may be drawn from this probability?

1 The difference is significant.
2 The difference is due to chance.
3 The difference is not due to chance.
4 The difference is due to some factor such as linkage of the genes concerned.

A 2 only **B** 3 and 4 only **C** 1, 2 and 4 only **D** 1, 3 and 4 only

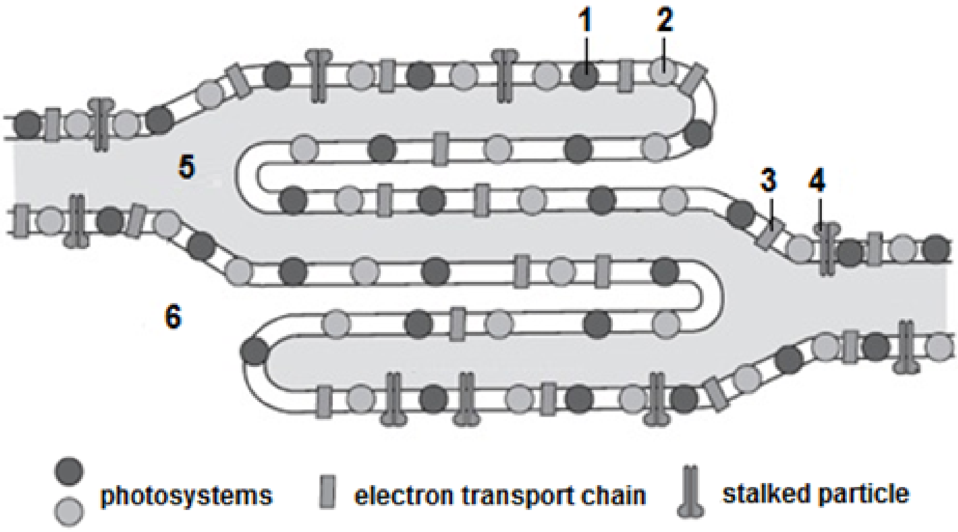
24 The diagram summarises the reactions during photosynthesis.



Which of the following correctly identifies the substances involved?

	1	2	3	4	5	6
A	H_2O	ADP, NAD	CO_2	O_2	ATP, NADH	glucose
B	O_2	ADP, NADP	glucose	H_2O	ATP, NADPH	CO_2
C	H_2O	ATP, NADPH	CO_2	O_2	ADP, NADP	glucose
D	H_2O	ADP, NADP	CO_2	O_2	ATP, NADPH	glucose

25 The figure shows the arrangement of photosystems, protein complexes containing chlorophyll molecules, on the thylakoid membrane of a plant chloroplast.



Which row correctly identifies the structure or function of the labelled regions?

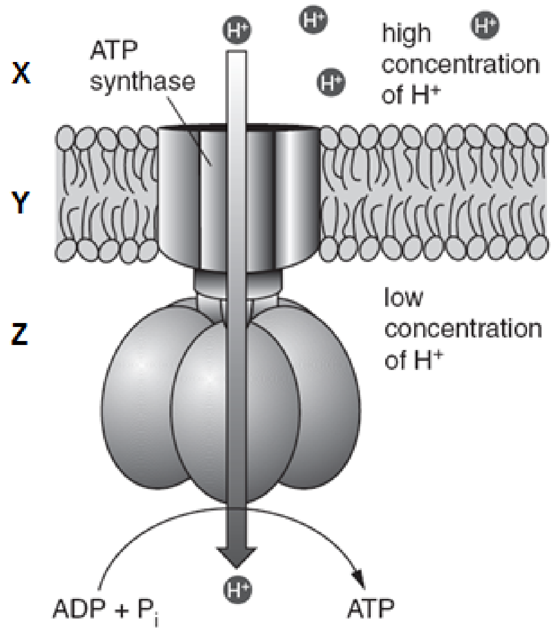
	site of photoactivation of chlorophyll	site of photolysis of water	site of ATP synthesis	stroma	thylakoid lumen
A	1 and 2	1 and 2	4	6	5
B	1 and 2	1	4	6	5
C	1, 2 and 3	1	3 and 4	5	6
D	1, 2 and 3	1 and 2	3 and 4	6	5

26 During substrate-level phosphorylation, ATP is synthesised from ADP and inorganic phosphate.

What is the immediate source of energy for this reaction?

A chemical bond energy released during the light independent stage of photosynthesis
B chemical bond energy released during glycolysis and the Krebs cycle
C potential energy of protons diffusing through the inner mitochondrial membranes in mitochondria
D potential energy of protons diffusing through thylakoid membranes in chloroplasts

27 The diagram shows part of a membrane in a cell.



Which of these descriptions is true?

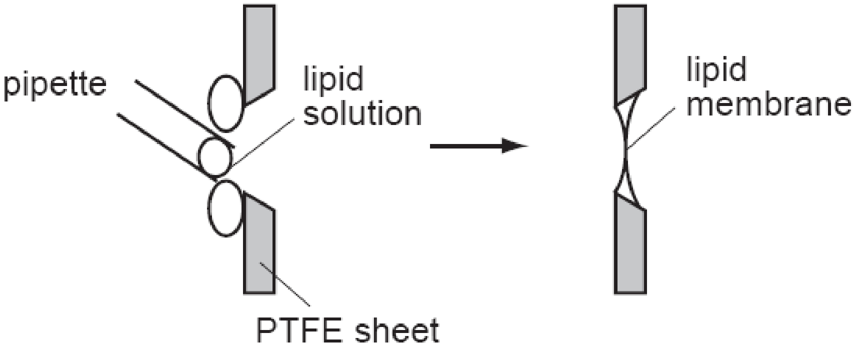
A X is the intermembrane space, Y is the inner mitochondrial membrane and the diagram shows ATP synthesis in a chloroplast.

B Y is the inner mitochondrial membrane, Z is the cytosol and the diagram shows ATP synthesis in a mitochondrion.

C X is the intermembrane space, Y is the inner mitochondrial membrane and the diagram shows ATP synthesis in a mitochondrion.

D Z is the intermembrane space, X is the matrix and the diagram shows ATP synthesis in a mitochondrion.

28 Lipid membranes can be formed in the laboratory by painting phospholipids over a PTFE sheet with a hole in it.



Such a lipid membrane is impermeable to water-soluble materials including charged ions such as Na^+ or K^+.

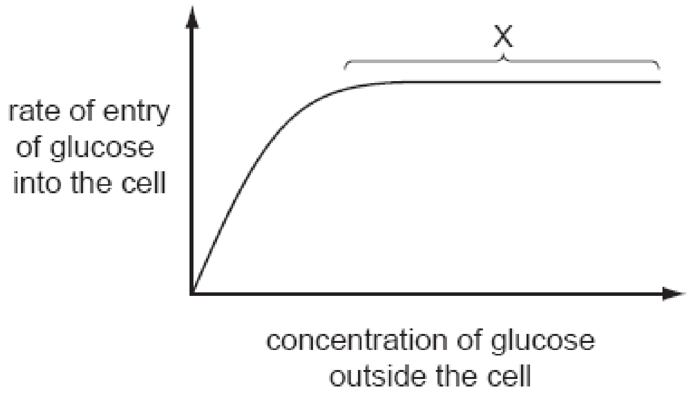
In one experiment with Na^+ ions, no current flowed across the membrane until a substance called gramicidin was added, at which time current flowed.

Which statement is consistent with this information and your knowledge of membrane structure?

Gramicidin becomes incorporated into the membrane and is

A a carbohydrate molecule found only on the outside of the membrane.
B a non-polar lipid which passes all the way through the membrane.
C a protein molecule with both hydrophilic and hydrophobic regions.
D a protein molecule which has only hydrophobic regions.

29 The graph shows how the rate of entry of glucose into a cell changes as the concentration of glucose outside the cell changes.



What is the cause of the plateau at X?

A All the carrier proteins are saturated with glucose.
B The carrier proteins are denatured and no longer able to function.
C The cell has used up its supply of ATP.
D The concentrations of glucose inside and outside the cell are equal.

30 Which two features contribute to the great tensile strength of cellulose?

1 glycosidic bonds linking the long chains of 1,4 α-glucose molecules
2 the -OH groups of the glucose molecules project outwards and form H bonds with neighbouring chains
3 the strength of the glycosidic bonds between the neighbouring chains of molecules
4 the successive glucose molecules are orientated at 180° to each other

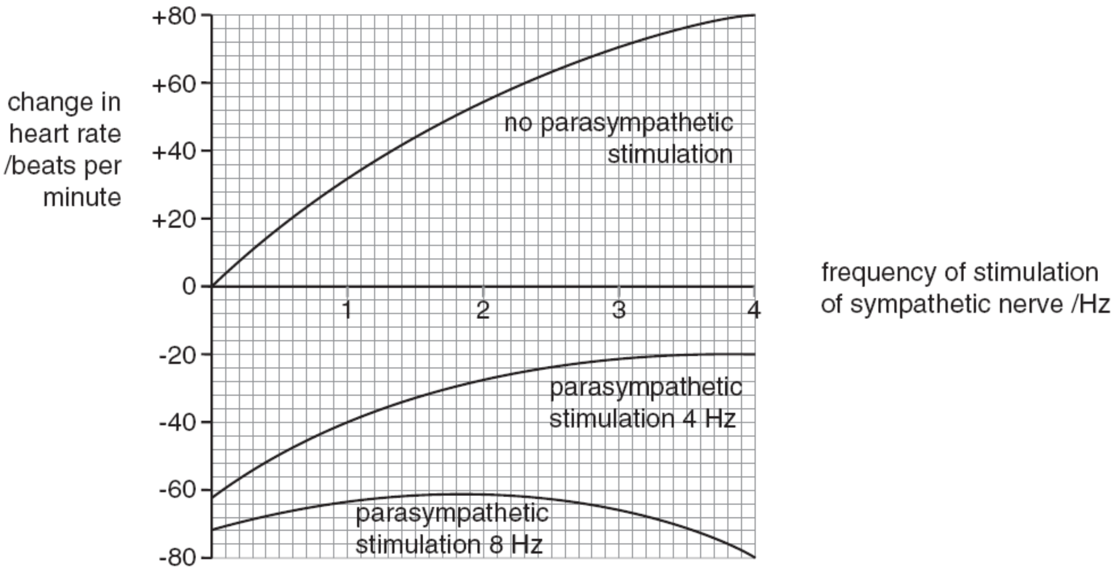
A 1 and 3 only **B** 1 and 4 only **C** 2 and 3 only **D** 2 and 4 only

31 During which stage of a nerve impulse does the opening of the voltage-gated sodium channels play an important role?

A resting potential
B depolarisation
C repolarisation
D hyperpolarisation

32 The heart, like most internal organs, is supplied with nerves from both the sympathetic and parasympathetic nervous systems. The neurones in these nerves form synapses with muscle tissue in the sino-atrial node (SAN) of the heart.

The diagram shows how the rate of heartbeat of a mammal changed when the sympathetic and parasympathetic nerves were stimulated at different frequencies. The frequency of stimulation was measured in hertz (Hz), i.e. the number of stimulations per second.



Which of these statements 1 to 4 are true?

1 The parasympathetic nerves released less transmitters substances than the sympathetic nerves.
2 The two nerves released different transmitter substances.
3 The stimulation of the sympathetic nerves decreases the heart rate while the stimulation of the parasympathetic nerves increases the heart rate.
4 The stimulation of the parasympathetic nerves decreases the heart rate while the stimulation of the sympathetic nerves increases the heart rate.

A 1 and 3 only **B** 1 and 4 only **C** 2 and 4 only **D** 1, 2 and 4 only

33 Which of the following is true about G-protein signalling?

1 During activation of G-protein, βγ subunit of the G-protein dissociates from the activated G-protein to activate adenylyl cyclase
2 During activation of G-protein, the active α subunit is terminated by the hydrolysis of the bound GTP caused by GTPase
3 Testosterone can bind to the cell membrane receptor to activate G-protein
4 The ratio of G-protein coupled receptor to G-protein is 1:1

A 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 2, 3 and 4 only

34 Before the settlement of California in the 1800s, the elk population was very large. By about 1900 there were only a few dozen elk left.

Owing to protection, there are now about 3000 elk living in a small number of isolated herds.

Unfortunately, some of the elk in all the herds have difficulty grazing due to a shortened lower jaw.

Which statements best explain this?

1 The early settlers only hunted elk that could graze.
2 There was a mutation affecting jaw size in one of the herds.
3 There is random mating within each herd.
4 The current elk population demonstrates a bottleneck effect.
5 There was directional selection favouring short jaws.

A 2 and 5 only
B 3 and 4 only
C 1, 2 and 4 only
D 2, 3 and 5 only

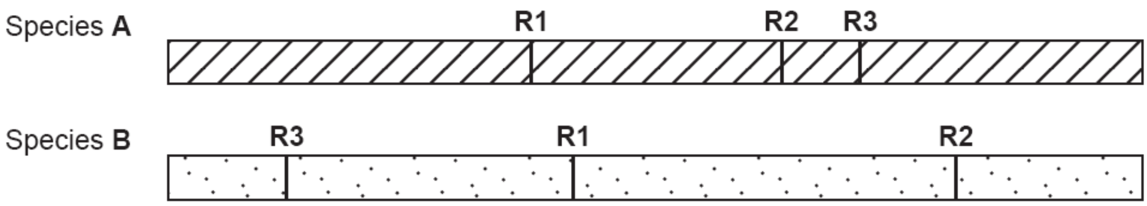
35 Darwin's view of the process of evolution to form new species (speciation) has been reinforced by more recent discoveries in genetics and cell biology.

In this view, which sequence of events is considered most likely to lead to speciation?

A	adaptation of population	→	competition and predation leading to natural selection	→	behavioural isolation	→	sympatric speciation
B	adaptation of population	→	competition and predation leading to natural selection	→	behavioural isolation	→	allopatric speciation
C	competition and predation leading to natural selection	→	geographical isolation	→	adaptation of isolated populations	→	sympatric speciation
D	competition and predation leading to natural selection	→	geographical isolation	→	adaptation of isolated populations	→	allopatric speciation

36 Restriction enzymes can be used to cut DNA at specific sites. Genes such as the gene for insulin can be cut from the chromosome of one species and as a result of ligation joined to the chromosome of another species forming recombinant (hybrid) DNA.

The figure shows two chromosomes from different species. The specific restriction enzyme sites (**R1**, **R2**, and **R3**) are shown.



Both chromosomes were cut at restriction site **R1** with *Eco*RI which recognises the sequence GAATTC and produces restriction fragments with sticky ends. The fragments were mixed and allowed to join to form recombinant (hybrid) DNA.

What is the maximum number of ways in which the fragments can join?

A 2 **B** 4 **C** 6 **D** 8

37 In which way(s) is/are the polymerase chain reaction (PCR) similar to the replication of DNA?

1 DNA is heated to break hydrogen bonds
2 DNA unwinds
3 RNA primers are used
4 DNA polymerase are required

A 2 only **B** 2 and 4 only **C** 1 and 3 only **D** 2, 3 and 4

38 Some of the steps involved in DNA fingerprinting are listed below:

1 transfer segments of DNA to nitrocellulose membrane
2 extraction of DNA
3 gel electrophoresis
4 treating DNA with restriction enzymes
5 autoradiography
6 hybridise with probe

The correct sequence is

A 2 → 4 → 3 → 1 → 6 → 5
B 4 → 2 → 3 → 1 → 6 → 5
C 4 → 2 → 1 → 3 → 6 → 5
D 2 → 4 → 3 → 1 → 5 → 6

39 A key feature of most multicellular organisms is the ability to differentiate and produce specialised cells.

Which row best describes the ability of zygotic cells to differentiate?

	totipotent	pluripotent	multipotent
A	✓	✓	✓
B	✓	✗	✓
C	✓	✗	✗
D	✗	✓	✓

Key: ✓ = ability, ✗ = no ability

40 Which row correctly states the effects and possible consequences of producing the various genetically-modified organisms?

	Bt corn		GM salmon		Golden rice	
	Effect	possible consequence	effect	possible consequence	effect	possible consequence
A	improve quality	might become weeds	improve yield	displace native species	improve yield	might become weeds
B	improve yield	toxic to closely related species	improve yield	displace native species	improve quality	cause allergies
C	improve yield	might become weeds	improve quality	cause allergies	improve yield	cause allergies
D	improve yield	toxic to closely related species	improve yield	cause allergies	improve quality	might become weeds